



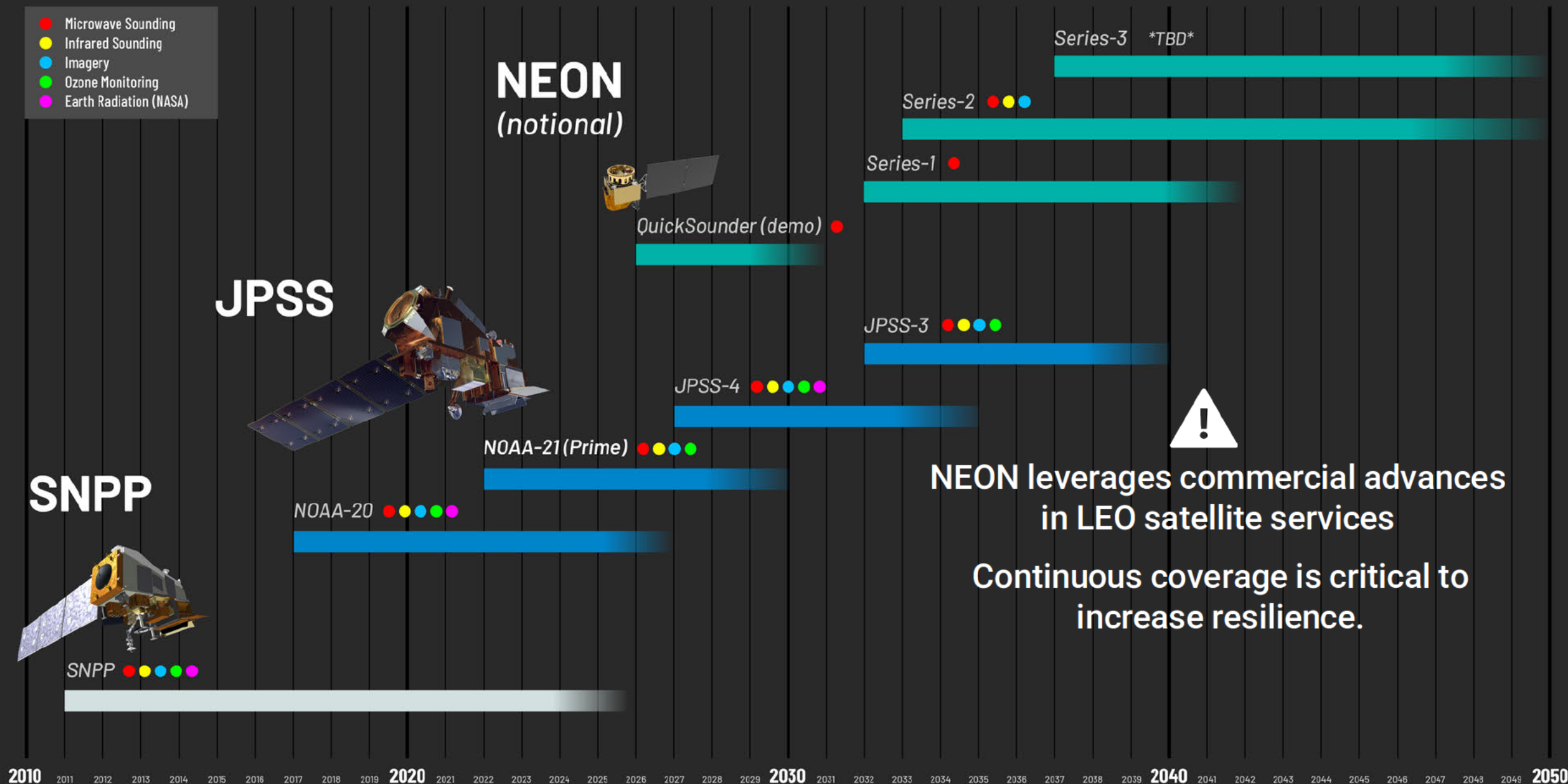
Streamlining Sounder for Microwave-Based Applications (SMBA) Development and Production

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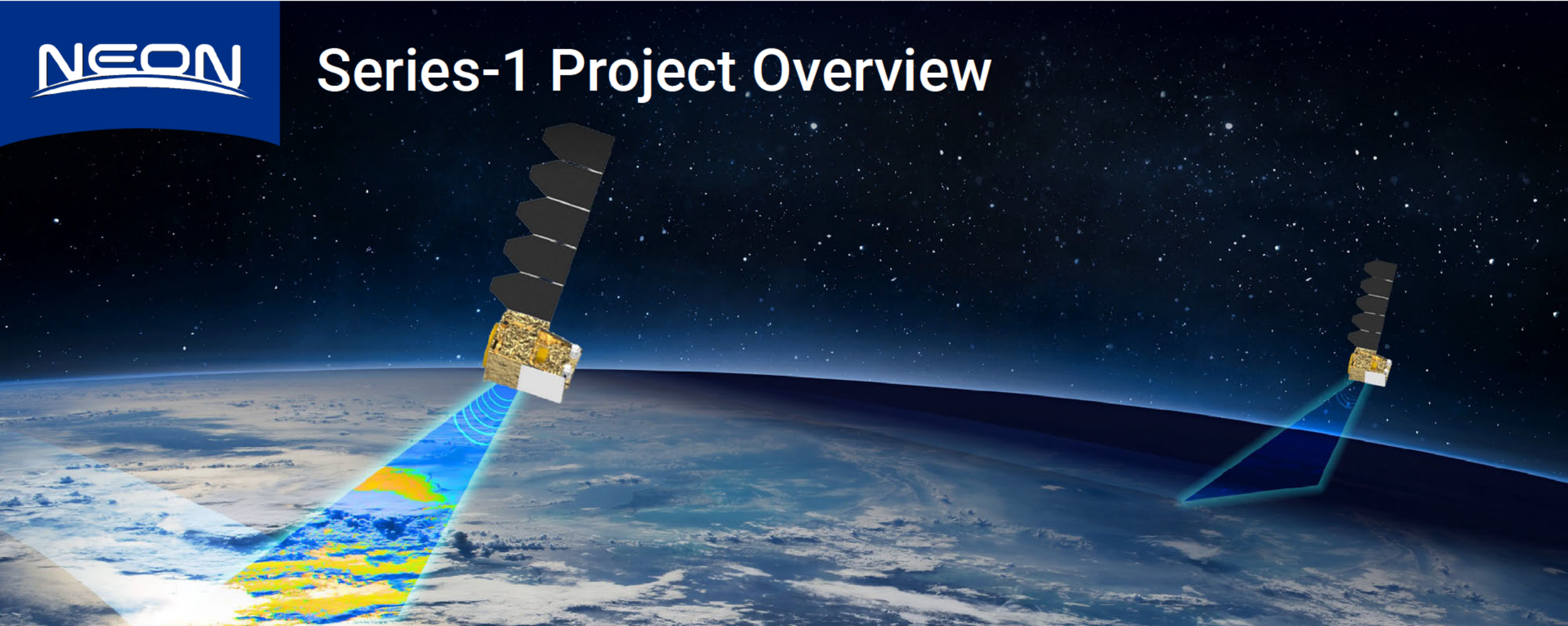


NOAA LEO Satellite Programs

- Microwave Sounding
- Infrared Sounding
- Imagery
- Ozone Monitoring
- Earth Radiation (NASA)



Series-1 Project Overview



Small satellites providing backbone microwave sounding capabilities essential to the performance of the NWP models.

INSTRUMENT Sounder for Microwave-Based Applications (SMBA)

ORBIT Polar-orbit sun-synchronous 13:30 LTAN

ALTITUDE 824 km

INCLINATION 98.7°

MISSION LIFE 5 years per satellite

LAUNCH AND OPERATIONS Commercially provided

- All architectural elements acquired commercially*!

- Instrument

} New Development

- Spacecraft

- Launch vehicle

- Mission operations

- Data Acquisition and Routing

} Bundled Mission Elements

SMBA needs a balanced approach to achieve next generation performance while leveraging savings from adopting non-Government processes and standards

**Note: This does not always imply FAR Part 12.*

What is Risk Posture?

C/o Mary Skow – NASA Agency Risk Management Officer

Definition

"An expression of the agreed upon limits of risk an organization's leadership team is willing to accept in order to achieve one or more of its objectives."

Purpose

Provides a consistent basis for decision-making and guides resource allocation for risk management.

Key Concept

Not about eliminating all risk, but defining what level of risk is acceptable in pursuit of objectives.

Essential Elements of a Well-defined Risk Posture

A. Set of Objectives – clearly defined set of organizational objectives, including programmatic constraints that provide the foundation for establishing a risk posture.

B. Risk Posture Statement – formally articulates the organization's acceptable risk limits in pursuit of its objectives. It aligns leadership on risk tolerance and guides decision-making at all levels.

C. Risk Tolerance Levels – defined boundaries that establish how much risk the organization is willing to accept for each objective. These tolerances guide decisions and help maintain consistency in risk management.

The Government intends to prioritize performance over process in accordance with the following prioritization approach:

*The SMBA Instrument prioritizes delivering heritage-equivalent (including ATMS, TROPICS, and MWS) performance and microwave sounding data continuity, delivering hyperspectral, RFI detection and intercalibration capabilities, **complying to science performance requirements** as specified in the PRD, and adhering to safety requirements. The SMBA instrument prioritizes **adopting non-Government processes and standards** that do not negatively impact science performance or safety to achieve cost-effective mission success. Where these areas conflict, **achieving SMBA performance and safety is the highest priority.***

This approach should be used for guidance in making informed decisions.

- Deferring to more industry processes
- Industry will use their established successful safety and mission assurance processes
 - Statement of Work and Instrument Mission Assurance Requirements combined into a single document
- Majority “Information” deliverables, some “Approval” deliverables, no “Review” deliverables
 - “Approval” deliverables typically associated with plans
- Deliverable Item Descriptions frequently allow best practices and contractor formats

- Performance requirements strongly reflect user community needs
- Relying on formal reviews, Engineering Peer Reviews (EPRs), contractor led meetings, and testing to provide confidence
 - Monthlies include more list type status items than before thus reducing overall deliverable requirements and formality
 - EPR plan identified critical subsystems for targeted Government involvement
 - Project chaired milestone reviews
- Establishing a trusting partnership that fosters transparency and technical insight
 - Enables fewer deliverables and fast tracks manufacturing

- Next Steps
 - Refine objectives
 - Refine relative priorities for each objective
 - Continue to balance priorities during implementation



SMBA will provide next generation performance while leveraging savings from adopting non-Government processes and standards

Back Up



- ARMO OneNASA Page (many links and resources are available here)
 - [Agency Risk Management Officer \(ARMO\)](#)
- Risk Management NASA Procedural Requirement (NPR 8000.4)
 - <https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=8000&s=4B>
- Enterprise Risk Management (Playbook and Greenbook)
 - <https://www.doi.gov/sites/doi.gov/files/erm-playbook-2022-update-final-508-compliant.pdf>
 - <https://www.gao.gov/assets/gao-14-704q.pdf>
- NASA's Risk Management Handbook (updated in 2025)
 - Part 1: <https://ntrs.nasa.gov/citations/20240014019>
 - Part 2: <https://ntrs.nasa.gov/citations/20240014326>
- Guidance on the Implementation of Objectives Driven, Risk Informed, and Case Assured Framework for S&MS (2025)
 - <https://ntrs.nasa.gov/citations/20250002210>
- Other helpful links
 - <https://sma.nasa.gov/sma-disciplines/risk-management>
- Risk Posture Statements
 - The University of Edinburgh: <https://information-services.ed.ac.uk/sites/default/files/2025-03/Risk%20Management%20Policy%20and%20Risk%20Appetite%20Statement.pdf>
 - World Food Program:
 - https://executiveboard.wfp.org/document_download/WFP-0000038870
 - https://executiveboard.wfp.org/document_download/WFP-0000099395